EXECUTIVE CONCEPT WHITE PAPER (DARPA STO / MTO / DSO & ONR GLOBAL BAA)

Native 16D Sedenion Signal Processing: Pre-LNA Passive RF Stripline S-Matrix, Pre-ADC Analog Crossbar, $K = 1$ Coherent Spoofer Detection, and Multi-Band Hardware

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Target Offices: DARPA STO (Assured PNT / EW / LPI-LPD) · DARPA MTO (Pre-ADC RF /RFSoC) · DARPA I2O/DSO (Lean 4 Assured Math) · ONR Global
Open-Source Code, RTL, SPICE & Lean 4 Proofs: https://github.com/josaum/sedenion-pg32 · DOIs: 10.5281/zenodo.23603006 & 10.5281/zenodo.23588274

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1. What are we trying to do? (Plain-Language Objective)
We aim to protect military and civilian RF/GNSS receivers and tactical LPI/LPD links from extreme electronic warfare (EW) jammers (+80 dB Jammer-to-Signal Ratio, JSR), co-linear mainbeam jammers ($\Delta\theta = 0.0°$), and zero-lag coherent repeater spoofers **before** front-end ADC saturation and **without** $\mathcal{O}(d^3)$ covariance matrix inversions ($\hat{\boldsymbol{R}}_{xx}^{-1}$). By mapping 16 real dual-polarized wavefield channels directly onto the 16-dimensional **sedenion algebra** ($\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8$) governed by PG(3, 2) and 4D Minkowski exterior calculus, our architecture realizes: (i) a **ferrite-free, passive, reciprocal 32-port RF stripline coupler** S**-matrix** that excises co-linear ($\Delta\theta = 0.0°$) linearly polarized jammers with 0.0000 dB **RHCP satellite insertion loss**; (ii) a **multiplier-free Pre-ADC Analog Sedenion Crossbar** (68.4–82.8 dB analog null depth); (iii) a $K = 1$ **single-snapshot associator curvature spoofer detector** (AUROC = 0.9999); (iv) a 512**-LUT**, 0**-DSP48**, 1**-cycle `Q1.14` RFSoC core**; and (v) **live multi-band bare-metal hardware reduction to practice** on `STM32WB55RG`, `ESP32-S2`, and ARM64 NEON, all backed by 170+ **machine-checked Lean 4 theorems** (`0 sorry`).

2. How is it done today, and what are the limits of current practice?
Current multi-antenna anti-jamming (MVDR / Capon STAP) and polarimetric array processors face four fundamental physical bottlenecks:

- **Pre-ADC Front-End Clipping & Ill-Conditioned $\mathcal{O}(d^3)$ Inversion:** Digital STAP/MVDR executes *after* the ADCs. At JSR = +80 dB, a 12-bit tactical ADC clips on 98.7% of samples ($\text{SINR}_{\text{post}} = -46.3$ dB, $P_d = 0.0\%$), while post-ADC $\hat{\boldsymbol{R}}_{xx}^{-1}$ requires 18,400 LUTs, 64 DSP48s, and 140 cycles on RFSoC and caps at 47.9 dB under 0.5° drift.
- **Mainbeam Collapse at Co-Linear Incidence ($\Delta\theta = 0.0°$):** When an airborne/UAV jammer aligns with the satellite line of sight ($\Delta\theta \to 0.0°$), spatial MVDR places a null directly on the satellite (SINR = −80.0 dB, GPS L1 C/A code-phase CRLB = 126,997 m).
- **Rank-1 Covariance Collapse Under Zero-Lag Coherent Spoofing:** A zero-lag repeater spoofer radiates the exact same scalar waveform $s(t)$ as the authentic emitter, collapsing the K-snapshot covariance $\hat{\boldsymbol{R}}_{xx}$ strictly to **Rank 1** for all $K \to \infty$ and blinding eigenvalue ratio detectors (AUROC = 0.2055, $P_{\text{det}} = 9.38\%$).
- **Hurwitz Division-Algebra Obstruction & RTK Cycle Slips:** Normed division algebras $(\mathbb{C}, \mathbb{H}, \mathbb{O})$ obey $\|u \cdot v\|_2 = \|u\|_2\|v\|_2$ and cannot zero an interferer via single-channel multiplication, while unconstrained MVDR weight updates induce PLL cycle slips ($> 15°$).

3. What is new in our approach, and why will it succeed?
Crossing the Hurwitz threshold to $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}e_8 \cong \mathbb{R}^{16}$ unlocks an 11D Stiefel cone of exact zero divisors $V_2(\mathbb{R}^7)$, an $8 + 7$ PG(3, 2) Fano hyperplane split, and 84 primitive zero-divisor blades $W_J = \frac{1}{\sqrt{2}}(e_c \pm e_d)$. We exploit five machine-checked hardware/algebraic breakthroughs:

1. **Pre-LNA Passive Reciprocal 32-Port RF Stripline S-Matrix & Co-Linear ($\Delta\theta = 0.0°$) Dual-Pol Jammer Excision:** Wiring RHCP $I/Q \to (e_1, e_2)$ and LHCP $I/Q \to (e_5, e_6)$ places every complex sample $X_J(\psi, s)$ of a tilt-ψ linearly polarized jammer on $V_2(\mathbb{R}^7)$ (`jam_linear_on_cone`). A single fixed zero divisor $v(\psi) = e_3 + \sin 2\psi\, e_{12} - \cos 2\psi\, e_{15}$ annihilates the *entire complex jammer waveform* (321.6 dB FP64; 79.8 dB at 10^{-4} coupler error) even at $\Delta\theta = 0.0°$, while passing the RHCP satellite at 0.0000 dB **distortionless insertion loss** and slashing the $\Delta\theta = 0.0°$ code-phase CRLB from **126,997** m → **17.96** m. Scaling $v_{\text{rf}}(\psi) = v(\psi)/2$ yields a **strictly passive ($\|S_{21}\|_2 \le 1$), reciprocal ($S_{32} = S_{32}^T$), zero ferrites,** 0 mW **DC)** 2-coupler-per-lane stripline network (`dualpol_tilt_passive_rf_stage`).
2. **Multiplier-Free Pre-ADC Analog Differential Sedenion Crossbar:** Because each of the 84 blades W_J has only 2 non-zero taps in $\{\pm 1\}$, we execute $Z(t) = R(t) \cdot W_J$ ahead of the ADCs using CMOS XOR muxes (`ADG1611`), DPDT ±1 wire-swaps, and ±0.05% thin-film summing (`AD8130` + `ERA-3AEB1001V`), achieving 68.38 dB (0.05%) **to** 82.76 dB (0.01% **foil) pre-ADC suppression** and restoring 100.0% **acquisition** (+21.28 dB **SINR)** on a 12-bit ADC at +80 dB JSR.
3. **Single-Snapshot ($K = 1$) Associator Curvature Spoofer Detector:** The native associator $\mathcal{A}_{\text{native}}(R) = [R, \pi_{\mathbb{O}}(R), e_8] = (-2(a \times_{\mathbb{O}} b), 0)$ satisfies $\|\mathcal{A}_{\text{native}}(R)\|_2^2 = 4(\|a\|_2^2\|b\|_2^2 - \langle a, b\rangle^2)$, achieving 99.80% **zero-lag**

spoofer detection (AUROC $= 0.9999$**) in** $K = 1$ **snapshot** and extracting $q_3 = a \times_{\mathbb{O}} b$ without SVD.

4. **Cycle-Slip-Free** PG$(3, 2)$ **Hinges,** $8 \oplus 4 \oplus 4$ **Transceiver & ZDKM LPI/LPD:** Routing carrier phase through the 3-node quaternion hinge $L_{12} = H_{\omega_1} \cap H_{\omega_2} \cong \mathbb{H}$ guarantees $[X_{\text{carrier}}, u_1, u_2]_{\mathbb{S}} \equiv 0$ ($0.0000°$ **phase jump**), while our co-designed $8 \oplus 4 \oplus 4$ transceiver ($J^2 = -4\boldsymbol{I}_4$) and ZDKM waveform achieve 99.90% keyed demodulation (49.76% radiometer intercept).

5. **Live Multi-Band Bare-Metal Hardware Validation (`STM32WB55RG`, `ESP32-S2`, ARM64 NEON):** Deployed in 1.4 KB Flash (0 B heap, 2.44 μs/call on NEON, 84.9 dB `Q1.14`) across 2,797 live Sub-GHz captures (**5.41**× jitter cut: 1.46 → 0.27 μs), 1,258 Optical IR bursts (**10.28**× jitter cut: 71.15 → 6.92 μs), 125 kHz LF-RFID (`0xFFFFFFFF` DMA underflow immunity, 99.41% PSK1), 13.56 MHz $K = 1$ cross-band rejection, and 2,738 live 2.4 GHz Wi-Fi trajectories.

Architecture / Subsystem	Domain / Algebra	Hardware Cost	Jammer Supp. ↑	$\Delta\theta = 0°$ CRLB ↓	L1 C/A P_d ↑	PLL Cycle-Slip ↓	$K = 1$ Spoof P_{det} ↑
`Oct-O2` / `Quat-H4` (Division)	$\mathbb{O}^2$ / $\mathbb{H}^4$	$\mathcal{O}(d^3)$ SVD	3.0–10.6 dB	126,997 m	0.0%	N/A	9.4% ($K = 64$)
`MVDR-SMI` (Post-ADC 12b / 64-Snap)	$\mathbb{C}^8$ ($\hat{\boldsymbol{R}}_{xx}^{-1}$)	18,400 LUT / 64 DSP48	0 dB (Clip) / 47.9 dB	126,997 m	0.0% / 8.2%	10.2% (5.80°)	9.4% (AUROC 0.21)
`LSMI-Loaded` ($K = 4$, $\delta = 10^{-2}$)	$\mathbb{C}^8$ ($\hat{\boldsymbol{R}}_{xx}^{-1}$)	18,400 LUT / 140 cyc	76.9 dB	126,997 m	30.5%	10.2%	9.4%
`NSSP Pre-LNA Stripline` ($v_{\text{rf}}(\psi)$)	$\mathbb{S}$ Passive S_{32}	**2 Couplers/Lane** (0 mW)	**79.8–321.6** dB	**17.96** m (**0.0** dB Loss)	**100.0**%	**0.0**% (**0.0000°**)	**99.8**% (**K = 1**)
`NSSP Pre-ADC Crossbar` (84-Blade)	$\mathbb{S}$ Analog XOR	**0 Mults** (±0.05% **R**)	**68.4–82.8** dB	**17.96** m	**100.0**% (+**21.3** dB)	**0.0**% (**0.0000°**)	**99.8**% (**K = 1**)
`Sed-PG32-Moreno` (`Q1.14` / FP64)	$\mathbb{S}/V_2(\mathbb{R}^7)$	**512 LUT / 0 DSP48 / 1 cyc**	**75.1–312.9** dB	**17.96** m	**100.0**% (+**21.5** dB)	**0.0**% (**0.0000°**)	**99.8**% (AUROC **1.0**)

4. High-Assurance Mathematical Verification in Lean 4 (`0 sorry`, 372 **Theorems)**

Every algebraic, electromagnetic, circuit-level, and bare-metal firmware invariant is machine-checked in **Lean 4** (`v4.15.0` + Mathlib, zero `sorry`, zero user axioms across `SedenionReal.lean`, `SedenionRobust.lean`, `Geometry.lean`, and `MaxwellExterior.lean`):

Assured EW / PNT / Hardware Guarantee	Lean 4 Theorem Identifier	Verification Domain	Proof Tactic
4D Maxwell Exterior Calculus	`d_sq_zero`, `poynting_stress_energy_dispersion`	4D Minkowski $\Omega^k(\mathbb{R}^{1,3})$	`ring` (`0 sorry`)
$K = 1$ Associator Wedge ($\|\mathcal{A}_{\text{native}}(R)\|_2^2 = 4(\|a\|^2\|b\|^2 - \langle a, b\rangle^2)$)	`native_associator_lagrange_norm`, `triad_orthogonality`	Continuous $\mathbb{R}^{16}$ (Mathlib)	`ring`
Co-Linear ($\Delta\theta = 0°$) Waveform Null & 0.0000 dB Sat Preservation	`dualpol_tilt_colinear_preservation`, `jam_linear_on_cone`	Continuous $\mathbb{R}^{16}$ (Mathlib)	`linear_combination`
Passive Reciprocal 32-Port RF Stripline S-Matrix ($\|S_{21}\|_2 \le 1$, $S_{32} = S_{32}^T$)	`dualpol_tilt_passive_rf_stage`, `right_mul_adjoint`	Continuous $\mathbb{R}^{16}$ (Mathlib)	`nlinarith` + `ring`
Co-Designed $8 \oplus 4 \oplus 4$ Multiplier-Free Transceiver ($J^2 = -4\boldsymbol{I}_4$)	`transceiver_joint_decode_and_null`	Continuous $\mathbb{R}^{16}$ (Mathlib)	`ring`
Zero-Cycle-Slip PLL Hinge & 16×16 Multi-Tile Kronecker Array (+6.02 dB)	`gnss_pll_hinge_associative`, `kronecker_multitile_16x16`	PG$(3, 2)$ & $\mathbb{S} \otimes \mathbb{S}$ ($\mathbb{Z}$)	`decide +kernel`
Bare-Metal `Q1.14` Exact 0 LSB Null, Glitch Coalescer & DMA Immunity	`q14_fixed_point_moreno_zero_lsb`, `lfrfid_hinge_arr_no_underflow`	Bare-Metal MCU ($\mathbb{Z}/\mathbb{N}$)	`decide +kernel` + `omega`

5. Who cares? If we are successful, what difference will it make? (DoD / Operational Payoff)

- **Surviving** +80 dB **JSR &** $\Delta\theta = 0.0°$ **Mainbeam Jamming Before the LNA/ADC:** Protects tactical SDRs and CRPAs from both ADC rail clipping (68.4–82.8 dB pre-ADC crossbar nulling) and co-linear mainbeam jammers (0.0000 dB RHCP satellite loss, 17.96 m CRLB vs. 126,997 m MVDR collapse) using passive 2-coupler-per-lane striplines (0 mW DC power, zero ferrites).
- 36× **Fewer FPGA LUTs, Zero DSP48 Multipliers, and** 140× **Lower Latency:** On Xilinx Zynq UltraScale+ RFSoC (`XCZU28DR`), our `Q1.14` core consumes **512 LUTs, 0 DSP48s, and 1 clock cycle** vs. 18,400 LUTs, 64 DSP48s, and 140 cycles for QR-SMI MVDR, while our bare-metal C/Rust engine runs in 1.4 KB Flash on $3 MCUs (`STM32WB55RG` / `ESP32-S2`).
- **Single-Snapshot (**$K = 1$**) Counter-Meaconing & Radiometer-Invisible LPI/LPD Comms:** Defeats Rank-1 zero-lag repeater spoofing in 1 sample (AUROC $= 0.9999$) and provides a white-covariance ($\boldsymbol{I}_{16}$) ZDKM waveform invisible to multi-antenna radiometers (49.76% intercept).

6. What are the technical risks and mitigation strategies?

1. **Risk 1 — Finite Jammer Axial Ratio, Antenna XPD, and Coupler Tolerance:** Imperfect linear jammer polarization (finite axial ratio AR) or coupler perturbation $\|\delta_v\|_2$ shifts the waveform slightly off $V_2(\mathbb{R}^7)$. *Mitigation:* Proven in `SedenionRobust.lean` (`leakage_bound` and `weight_quantization_leakage_bound`), residual leakage obeys $\|u \cdot \tilde{v}\|_2^2 \le 2\|u\|_2^2\|\delta_v\|_2^2$: at AR = 40 dB (37.0 dB RF stripline suppression) and XPD = 30 dB (0.076 dB satellite loss), cascading the Pre-LNA Stripline Stage with the Pre-ADC Crossbar (68.4 dB) exceeds > 105 dB combined suppression.
2. **Risk 2 — Scaling Beyond 4 Simultaneous Jammers / 16 Channels:** A single 16D sedenion block has $\dim\ker L_{\mathbb{S}}(u) = 4$ (313.3 dB verified across $M = 4$ simultaneous jammers). *Mitigation:* Our machine-checked 16×16 Multi-Tile Kronecker Sedenion Array ($\mathbb{S} \otimes \mathbb{S} \cong \mathbb{R}^{256}$, Theorem 5.21 in `MaxwellExterior.lean`) cascades two 16D stages with +6.02 dB (16× power) gain and zero matrix inversions.

7. Cost, Schedule, and Mid-Term / Final Verification Exams (12-Month Seedling Plan — $485k)

We propose a **12-month, $485k Seedling effort** (TRL 4 → TRL 6) structured around two quantitative verification gates:

Phase / Timeline	Budget	Technical Deliverable	Quantitative Verification Exam ("Gate")
Phase I (Months 1–6)	\$240k	Fabricate the 4-layer Rogers `RO4350B` Dyadic 2×2 Dual-Pol TEM Array, Passive Stripline S_{32} Coupler Feed, and 16-Ch Pre-ADC Analog Crossbar Mezzanine (`ADG1611`/`AD8130`).	**Mid-Term Exam (Month 6):** ≥ 65 dB pre-ADC analog jammer suppression prior to 12-bit ADC quantization, < 0.25 dB RHCP satellite loss at $\Delta\theta = 0°$, and 0% ADC clipping at +80 dB JSR.
Phase II (Months 7–12)	\$245k	Integrate Xilinx RFSoC (`XCZU28DR`) 512-LUT `Q1.14` core + $K = 1$ Associator Spoofer Detector and execute over-the-air ZDKM LPI/LPD & +80 dB JSR HIL trials.	**Final Exam (Month 12):** 100% GPS L1 C/A acquisition at +80 dB JSR, $> 99\%$ single-snapshot ($K = 1$) zero-lag repeater spoofer detection, and BER $< 10^{-3}$ ZDKM LPI/LPD link.

8. Immediate Reproducibility (Available Today for DARPA / ONR Evaluators)
All Lean 4 proofs (372 theorems across `lean/` and `MaxwellExterior.lean`, `0 sorry`), SPICE Monte Carlo suites, SystemVerilog RTL, and benchmark scripts execute in < 2 minutes:

- **Zenodo DOIs & GitHub:** 10.5281/zenodo.23588274 & 10.5281/zenodo.23603006 — https://github.com/josaum/sedenion-pg32
- **Benchmarks & Lean 4 Kernel:** `python3 benchmarks/gps_jamming/run_native_sedenion_dsp.py`, `run_dualpol_pivo` `run_phase2_defense.py`, and `lake build && lake env lean Axioms.lean`.